Curriculum Vitae
Oct 26

Alen E. Golpashin

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EDUCATION

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

Ph.D. Aerospace Engineering — Aug 21 - Dec 26 (Expected)
Adviser: Bruce A. Conway
Dissertation title: *Dimensionality in Minimum-Time Bellman Equations*

M.S. Aerospace Engineering — Aug 16 - May 18
Advisers: N. Sri Namachchivaya, Koki Ho

B.S. Aerospace Engineering — Aug 13 - May 16

LOYOLA UNIVERSITY CHICAGO

M.S. Mathematics — Aug 18 - Dec 19

RESEARCH INTERESTS

Optimal control, dynamical systems, Hamilton–Jacobi equations, numerical methods for PDEs, homogenization and dimension reduction, reinforcement learning, data assimilation and filtering.

PUBLICATIONS

- Golpashin A.E., Clark M.T., and Conway B.A., "Minimum-Time Feedback Transfers and Reachability in the Planar Circular Restricted Three-Body Problem Using Viscosity-Informed Learning," To be presented at the *37th AAS/AIAA Space Flight Mechanics Meeting*, New Orleans, Louisiana, January 17–21, 2027. [View]
- Golpashin A.E., Puthumanaillam G., Ornik M., and Conway B.A., "Viscosity-Informed Generative Actor-Critic for High-Dimensional Stochastic Optimal Control," ArXiv preprint, arXiv:2605.15668, 2026. [View]
- Golpashin, A.E., Clark, M.T., and Conway, B.A. "Adaptive Grid Refinement for Optimal Feedback Control." *47th Annual AAS Guidance, Navigation, and Control Conference*, Breckenridge, Colorado, January 31–February 5, 2025. [View]
- Conway, B.A., Chilan, C.M., Golpashin, A.E., "Optimal Feedback Control of Astrodynamic Systems Using Solutions of the Hamilton-Jacobi-Bellman Equation,". In: Cannarsa, P., Celletti, A., Fasano, G., Mazzini, L., Pontani, M., Trélat, E. (eds) *New Trends and Challenges in Optimization Theory Applied to Space Engineering*. Springer Optimization and Its Applications, vol 221. pp. 161–183. Springer, Cham, 2025. [View]
- Golpashin A.E., Yeong H.C., Ho K., and Sri Namachchivaya N., "Spacecraft Attitude Control: A Consideration of Thrust Uncertainty," *Journal of Guidance, Control, and Dynamics*, 2020. [View]
- Golpashin, A. E., "Hamilton-Jacobi-Bellman equation for stochastic optimal control: applications to spacecraft attitude control," Master's Thesis, Department of Aerospace Engineering, University of Illinois, Urbana, IL, 2018. [View]
- Golpashin A., Yeong H.C., Ho K., and Sri Namachchivaya N., "Stochastic Attitude Control of Spacecraft under Thrust Uncertainty," *AIAA/AAS Astrodynamics Specialist Conference,* Stevenson, WA, Aug. 2017. [View]

IN PROGRESS WORKS

- "Homogenization in Minimum-Time Control,"
- "Finite-Horizon Robotic Control with Time-Dependent Viscosity-Informed Learning,"
- "Detection of Volumetric Cyber Attacks using Kalman-Lévy Filter," [Summary, Preprint]

RESEARCH EXPERIENCE

Conway Research Group (Dept. of Aerospace Engineering) — Urbana, IL
Graduate Researcher — Aug 20 - Dec 26

Nonlinear Mechanics and Dynamics Research Institute (Sandia National Labs) — Albuquerque, NM
Graduate Researcher — May 23 - Aug 23

Center for Cyber Warfare (Naval Postgraduate School) — Monterey, CA
ORISE Graduate Research Intern — May 21 - Aug 21

Nonlinear Systems Group (Dept. of Aerospace Engineering) — Urbana, IL
Graduate Researcher — May 16 - May 20

Space Systems Optimization Laboratory (Dept. of Aerospace Engineering) — Urbana, IL
Graduate Researcher — May 16 - May 20

Bretl Research Group (Coordinated Science Laboratory) — Urbana, IL
Undergraduate Research Assistant — Jan 15 - May 16

TEACHING AND MENTORING

Education Justice Project – For Credit Course at Danville Correctional Center — Danville, IL
Course Instructor (AE 353 – Aerospace Control Systems) — Aug 23 - Dec 23
Department of Mathematics and Statistics – Loyola University Chicago — Chicago, IL
Graduate Teaching Assistant
Courses
Math 263 – Multivariable Calculus (2 sections) — Aug 19 - Dec 19
Math 264 – Ordinary Differential Equations — Jan 19 - May19
Math 212 – Linear Algebra — Jan 19 - May19
Math 264 – Ordinary Differential Equations — Aug 18 - Dec 18
Math 212 – Linear Algebra — Aug 18 - Dec 18
Department of Aerospace Engineering – University of Illinois — Urbana, IL
Graduate Teaching Assistant
Courses
AE 483 – Unmanned Aerial Vehicle (UAV) Navigation and Control (Hardware lab with Crazyflie 2.0) — Aug 22 – Dec 22
AE 352 – Aerospace Dynamical Systems — Jan 18 - May18
AE 202 – Aerospace Flight Mechanics — Aug 17 - Dec 17
AE 352 – Aerospace Dynamical Systems — Jan 17 - May 17
AE 433 – Aerospace Propulsion — Aug 16 - Dec 16
Writers Workshop — Urbana, IL
Science Consultant — Jan 23 - May 23
Research Seminar Mentoring (AE 298) — Urbana, IL
Graduate Mentor — Aug 22 - May 23
Illinois Aerospace Institute (High School Summer Camp) — Urbana, IL
Orbital Mechanics Instructor — Summer 2018
Mentoring Undergraduates in Science and Engineering (MUSE) Program — Urbana, IL
Graduate Mentor — Apr 17 - May 17
Graduate Mentor at Aerospace Engineering Department — Urbana, IL
AE 397 – Independent Study (with 1 undergraduate student) — Jan 17 - Dec 17
AE 497 – Independent Study (with 2 undergraduate students, 1 student continuing from AE 298) — Aug 23 - Present
Office of Minority Student Affairs — Urbana, IL
Undergraduate (Math and Physics) Tutor — Jan 14 - May 15
Mathematical Sciences Learning Center — Chicago, IL
Undergraduate Mathematics Tutor — Aug 11 - May 13

TALKS AND PRESENTATIONS

- *Tackling the Dimensionality Challenge in HJB PDEs: A Reinforcement Learning Approach to Adaptive Grid Refinement*
  2025 American Mathematical Society Spring Central Sectional Meeting, University of Kansas, KS, March 2025. [View]
- *Analytical Modeling of Piezoelectric Stack Actuators for Vibrafuge Applications*
  2023 NOMAD Research Institute, University of New Mexico, Albuquerque, NM, August 2023. [View]
- *A Study of Non-Gaussian Data Assimilation for Volumetric Network Anomaly Detection*
  2021 ORISE project presentations, Naval Postgraduate School, Monterey, CA, August 2021. [View]
- *Stochastic Optimal Control in Spacecraft Attitude Control Applications*
  12th Annual Loyola Graduate School Interdisciplinary Research Symposium, Chicago, IL, March 2019. [View]
- *Optimized Rendezvous of a Quadrotor*
  Aerospace Engineering Undergraduate Research Poster Session, Urbana, IL, May 2016. [View]

FELLOWSHIPS AND FUNDING

Graduate Assistance in Areas of National Need (GAANN) Fellowship - U.S. Department of Education — Urbana, IL
Awardee — Aug 20 - May 22
Grant FA8651-23-1-0001 - Air Force Research Laboratory (AFRL) — Urbana, IL
Proposal coauthor and graduate researcher — Aug 23 - Aug 26

UNIVERSITY SERVICE AND ACTIVITIES

Graduate Student Advisory Committee (AeroGSAC) Urbana, IL
President Aug 23 - Aug 24
Outreach & Advocacy Subcommittee Chair Aug 22 - Aug 23
Interdepartmental Subcommittee Chair Aug 17 - May 18
- Organized "Navigating Graduate School and Beyond," a virtual Q&A session with alumni Apr 24
- Organized a virtual Q&A session with alumni on industry careers Feb 24
- Organized a high school visitation program for prospective students in collaboration with SHPE Oct 22
- Co-organized "Aerospace Engineering Graduate Research Symposium", with Rolls-Royce's sponsorship May 18

Engineering Graduate Student Advisory Committee (EGSAC) Urbana, IL
Member Aug 22 - May 23
Illinois Robotics in Space (IRIS) Urbana, IL
Drive System Designer Aug 15 - May 16
Research Journal Peer Review
Single-anonymous peer reviewer
- Journal of Guidance, Control, and Dynamics (JGCD) 20
- Journal of Optimization Theory and Applications (JOTA) 22

INDUSTRY EXPERIENCE

HydraForce Inc., Innovation and Technology Center Vernon Hills, IL
Design Engineering Intern May 14 - Aug 14
HydraForce Inc., Headquarters Lincolnshire, IL
Manufacturing Engineering Intern May 13 - Aug 13
CITES Classroom & Conference Media Engineering Urbana, IL
Classroom Tech Support Jan 14 - Jan 15

SELECTED COURSEWORK

Theoretical Courses: Real analysis I & II, Measure theory, Complex analysis, Functional analysis, Abstract algebra I & II (group theory, ring theory, and introductory homological algebra), Abstract linear algebra, Dynamical systems and bifurcation theory, Discrete dynamical systems and chaos theory, Applied dynamical systems, Optimization (linear programming), Stochastic processes, Partial differential equations.
Applied Courses: Systems dynamics & control, Spacecraft attitude dynamics and control, Intro to robotics, Optimal aerospace systems, Robust control, Distributed system control, Analog signal processing, Aeroelasticity, Geometric control theory, Stochastic control theory, Reinforcement learning, Optimal Spacecraft Trajectories.